

A sharp regularity classification for countable exponential-feedback transseries

Convergence, exact Gevrey type, coefficient condensation,
and sectorial realization

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Abstract

We study the formally well-defined equation

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q)), \quad \lambda_j \geq 0,$$

as a model of countably many exponentially delayed corrections in transseries inversion. Formal existence alone imposes no growth restriction on the slopes λ_j . We derive a sharp separation between formal, analytic, and Gevrey regularity. The solution has positive radius of convergence exactly when $\lambda_j = O(j)$. More precisely, writing $\widehat{U} = \sum_{n \geq 1} u_n q^n$, for every $s > 0$ we prove the exact identity

$$\limsup_{n \rightarrow \infty} \left(\frac{u_n}{(n!)^s} \right)^{1/n} = \left(\frac{s+1}{s} \right)^{s+1} \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^{s+1}}.$$

For $\lambda_j \sim a j^p (\log j)^\beta$, $p > 1$, we determine $\log u_n$ through its term of order n , and prove concentration of the Lagrange weights on configurations with one large action of size $\frac{p}{p-1} \frac{n}{\log n}$.

The quadratic model has an entire ordinary Borel transform with $\log \log \mathcal{B}\widehat{U}(t) \sim 2\sqrt{t}$, hence is not Borel–Laplace summable on the positive ray. Nevertheless, every nonnegative slope sequence, even one of superpolynomial growth, admits a holomorphic solution with this Poincaré expansion on proper subsectors of the left half-plane. We give explicit action-truncation bounds, transfer the Gevrey classification to formal compositional inverses, and provide exact-arithmetic checks and a focused research program.

Status and scope. The results are proved here as conventional mathematical statements, not as Lean theorems. The sharp classification and its quantitative consequences are the proposed contribution of this research draft. Lagrange inversion, Gevrey closure under reversion, and contraction arguments are classical methods, not claimed inventions. A priority claim against the entire literature is not established, and no named longstanding conjecture is claimed solved.

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1 Problem selection and connection with ProveIt

1.1 The repository boundary being investigated

The ProveIt repository's consolidated *Transseries: the polynomial-logarithmic calculus and its inversions* connects formal asymptotic scales, composition and reversal, Lambert-type equations, analytic realization, and remainder certificates. Its README explicitly separates the human mathematics from the scope of the existing Lean APIs [1]. In particular, formal support control and analytic realization are distinct tasks. This paper investigates that distinction for a concrete countable implicit kernel, rather than adding another finite list of Lambert coefficients.

The repository snapshot consulted is commit `04e06e032dff1966513bfba316966d52db3646b4`. The relevant directory is

```
Analysis/Transseries/docs/  
series-and-transseries/Transseries_And_Inversion/.
```

Editorial note (ProveIt, 2026-09-29). The directory is given at its current location. At the pinned snapshot it was

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/  
series-and-transseries/Transseries_And_Inversion/;
```

the transseries material has since moved to `Analysis/Transseries/` (see `docs/incoming/README.md`, batch 45). Packages filed later beside this one in `Analysis/Transseries/docs/series-and-transseries/` answer several questions of Section 13; see the editorial notes there.

We use its documented program as motivation. None of the proofs below requires accepting an unverified frontier theorem from that repository. Nor do we assert that the particular classification problem formulated here was already posed there verbatim.

The central question is the following:

When countably many formally admissible exponential corrections feed back into an implicit inverse, what growth of their feedback slopes is compatible with convergence, with a specified Gevrey class, and with an actual sectorial solution?

A single-grid model already exhibits all three phenomena. Thus the obstruction does not depend on exotic supports or a failure of Hahn summability.

1.2 The model and its transseries interpretation

Fix any sequence of finite real numbers $\lambda_j \geq 0$. Put

$$\Phi_{\lambda}(q, u) = \sum_{j \geq 1} q^j e^{\lambda_j u}, \quad \widehat{U} = \Phi_{\lambda}(q, \widehat{U}). \quad (1.1)$$

Initially this is a *formal* identity. No analytic convergence of Φ_{λ} near $(0, 0)$ is assumed. On writing $q = e^{-x}$, the solution becomes

$$\widehat{U}(e^{-x}) = \sum_{n \geq 1} u_n e^{-nx}. \quad (1.2)$$

This is a well-based, one-generator exponential transseries. Each primitive term has action j and feedback slope λ_j . Formal transseries allow arbitrary real coefficients on such a support; coefficient growth is a separate analytic question. This distinction is part of the general formal framework described by van der Hoeven [2].

The model is a nonlinear implicit inversion problem: one solves $u - \Phi_{\lambda}(q, u) = 0$ for u , and the resulting tangent-to-identity series itself has a formal compositional inverse. Section 10 transfers the classification to that inverse.

1.3 What is classical and what is obtained here

The coefficient formula in Section 3 is an application of classical Lagrange inversion [3]. Infinite-colour inversion and convergence theorems exist in much more general settings, including the work of Jansen, Kuna, and Tsagkarogiannis [4, 5]. We do not claim a first infinite-variable inversion theorem.

The specific contribution is the exact relationship between slope growth and coefficient growth for (1.1), including the constant in the Gevrey-type identity. Its proof is an elementary but quantitative optimization of the positive Lagrange formula. The same argument produces logarithmic coefficient asymptotics and a concentration theorem. A second, independent construction realizes the formal solution in left-hand sectors without any slope-growth hypothesis.

Existing Gevrey implicit-function theorems require analytic or sectorial uniformity hypotheses on the defining equation. For example, Nikolaev's theorem assumes suitable uniform Gevrey asymptotics in a genuine neighborhood of the dependent variable [6]. Our rapidly growing slopes prevent such a neighborhood at $(0, 0)$. The counterexamples below therefore do not contradict those theorems. Likewise, an entire Borel transform is not by itself a Borel–Laplace summation theorem: growth at infinity must also be controlled [7].

2 Main results and conventions

Throughout, $\log$ denotes the natural logarithm, $0^0 = 1$ in coefficient formulas, and $\mathbb{N}$ includes zero. All limits involving $\log j$ start at $j \geq 2$. A multi-index $\mathbf{m} = (m_1, m_2, \dots)$ always has finite support. Write

$$|\mathbf{m}| = \sum_j m_j, \quad w(\mathbf{m}) = \sum_j j m_j, \quad \lambda \cdot \mathbf{m} = \sum_j \lambda_j m_j, \quad \mathbf{m}! = \prod_j m_j!.$$

Definition 2.1. For $s \geq 0$, a formal series $\hat{f} = \sum f_n q^n$ is in $\mathcal{G}_s$ if there are $C, A > 0$ such that $|f_n| \leq C A^n (n!)^s$ for every n . For $s > 0$ its factorial-normalized type is

$$T_s(\hat{f}) = \limsup_{n \rightarrow \infty} \left(\frac{|f_n|}{(n!)^s} \right)^{1/n}.$$

Types may be zero or infinite. The class $\mathcal{G}_0$ is precisely the class of convergent power series.

Theorem 2.2 (Sharp regularity classification). *Equation (1.1) has a unique solution $\hat{U} \in q\mathbb{R}[[q]]$, with $u_1 = 1$ and $u_n \geq 1$. Its radius of convergence is positive if and only if*

$$\sup_{j \geq 1} \frac{\lambda_j}{j} < \infty. \quad (2.1)$$

For every $s > 0$, the following exact identity holds in $[0, \infty]$:

$$\boxed{T_s(\hat{U}) = C_s A_s(\boldsymbol{\lambda}), \quad C_s = \left(\frac{s+1}{s} \right)^{s+1}, \quad A_s(\boldsymbol{\lambda}) = \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^{s+1}}.} \quad (2.2)$$

Consequently,

$$\hat{U} \in \mathcal{G}_s \iff \lambda_j = O\left((j \log(j+1))^{s+1}\right), \quad s > 0. \quad (2.3)$$

There is a real endpoint distinction: (2.3) must not be extrapolated to $s = 0$. For example, $\lambda_j = j \log(j+1)$ gives a divergent series that belongs to every positive Gevrey class.

Theorem 2.3 (Regularly varying slopes). *Suppose, for constants $a > 0$, $p > 1$, and $\beta \in \mathbb{R}$, that*

$$\lambda_j \sim aj^p(\log j)^\beta.$$

Then, as $n \rightarrow \infty$,

$$\begin{aligned} \log u_n &= (p-1)n \log n + (\beta-p)n \log \log n \\ &\quad + \left[\log a + p \log \frac{p}{p-1} - (p-1) \right] n + o(n). \end{aligned} \quad (2.4)$$

Under the probability distribution on $w(\mathbf{m}) = n$ proportional to

$$W(\mathbf{m}) = \frac{(\lambda \cdot \mathbf{m})^{|\mathbf{m}|-1}}{\mathbf{m}!},$$

let $K = |\mathbf{m}|$, $R_n = n - K + 1$, and $J_{\max} = \max\{j : m_j > 0\}$. Then

$$\frac{R_n \log n}{n} \rightarrow \frac{p}{p-1}, \quad \frac{J_{\max}}{R_n} \rightarrow 1 \quad (2.5)$$

in probability. Each fixed positive deviation in either statement has probability bounded by $\exp(-cn + o(n))$ for some $c > 0$.

The concentration statement concerns precisely this finite Lagrange-weight probability space. It is not a blanket assertion about all models of random rooted trees.

Theorem 2.4 (An entire Borel transform with excessive growth). *For $\lambda_j \sim aj^p$, $p > 1$, set $s = p - 1$ and define*

$$B_s(t) = \sum_{n \geq 1} \frac{u_n}{(n!)^s} t^n.$$

Then B_s is entire and

$$\lim_{t \rightarrow +\infty} \frac{\log \log B_s(t)}{t^{1/p}} = a^{1/p} \frac{p}{p-1}. \quad (2.6)$$

In particular, for $\lambda_j = j^2$ the ordinary integrated Borel transform B_1 satisfies $\log \log B_1(t) \sim 2\sqrt{t}$. The ordinary positive-ray Laplace integral of B_1 diverges for every positive value of q .

Theorem 2.5 (Universal left-sector realization). *For every sequence $\lambda_j \geq 0$ and every $\eta \in (0, 1)$, there is $r_0 > 0$ such that the sector*

$$S_{\eta, r_0} = \{q : 0 < |q| < r_0, \Re q < -\eta|q|\}$$

supports a holomorphic solution U of the literal, absolutely convergent identity $U = \Phi_\lambda(q, U)$, with $U = q + O(q^2)$. It has Poincaré expansion $\widehat{U}$ on that sector. Its finite-action approximations satisfy an explicit error bound of order $|q|^{N+1}$, uniformly in the number N of retained actions.

Theorems 2.2–2.5 address different questions. Sectorial existence does not assert convergence at the origin; Poincaré asymptotics does not assert uniform Gevrey remainder bounds; and an entire Borel transform does not assert adequate Laplace growth.

3 Formal existence and the positive coefficient formula

Lemma 3.1 (Admissibility and coefficientwise contraction). *For any $u \in q\mathbb{R}[[q]]$, the right-hand side of (1.1) is a well-defined element of $q\mathbb{R}[[q]]$. If $u - v$ has order $m \geq 1$, then*

$$\text{ord}_q(\Phi_\lambda(q, u) - \Phi_\lambda(q, v)) \geq m + 1.$$

Consequently the iterates $U^{[0]} = 0$ and $U^{[h+1]} = \Phi_\lambda(q, U^{[h]})$ stabilize coefficientwise, with the coefficient of q^n stable once $h \geq n$.

Proof. At coefficient q^n , only $j \leq n$ can occur. In the exponential attached to such a j , only powers u^k with $k \leq n - j$ contribute. Thus every coefficient is a finite sum. The formal difference $e^{\lambda_j u} - e^{\lambda_j v}$ is divisible by $u - v$; multiplication by q^j raises order by at least one. This proves the contraction estimate. Completeness of $\mathbb{R}[[q]]$ in the coefficientwise, or q -adic, topology gives a fixed point. If two fixed points differed first in degree m , the same estimate would make their difference have order at least $m + 1$, a contradiction. Iteration also proves the stabilization assertion. $\square$

Proposition 3.2 (Two equivalent Lagrange formulas). *The coefficients of $\hat{U}$ are*

$$u_n = \sum_{\substack{\mathbf{m} \geq 0 \\ w(\mathbf{m})=n}} \frac{(\lambda \cdot \mathbf{m})^{|\mathbf{m}|-1}}{\mathbf{m}!}, \quad (3.1)$$

$$= \sum_{k=1}^n \frac{1}{k!} \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} (\lambda_{j_1} + \dots + \lambda_{j_k})^{k-1}. \quad (3.2)$$

In particular, $u_n \geq 1$, and each u_n is nondecreasing in every slope.

Proof. Introduce a marker t for the number of primitive factors and solve $V = t\Phi_\lambda(q, V)$. Classical formal Lagrange inversion [3] gives

$$[t^k]V = \frac{1}{k} [z^{k-1}] \Phi_\lambda(q, z)^k.$$

For the ordered tuple $(j_1, \dots, j_k)$, the product of exponentials is $\exp((\sum_i \lambda_{j_i})z)$, whose coefficient of z^{k-1} is $(\sum_i \lambda_{j_i})^{k-1}/(k-1)!$. This proves (3.2). Setting $t = 1$ is legitimate coefficientwise because $k \leq n$ in degree q^n . Grouping ordered tuples by their multiplicities gives $k!/|\mathbf{m}|!$ tuples in each group and proves (3.1). The tuple (n) contributes one, and all terms are nonnegative. Monotonicity is immediate. $\square$

3.1 A useful lower bound

For integers $r \geq 2$ and $k \geq 1$ with $r + k - 1 = n$, retain the multiplicities $m_r = 1$, $m_1 = k - 1$. Equation (3.1) yields

$$u_n \geq \frac{(\lambda_r + (k-1)\lambda_1)^{k-1}}{(k-1)!} \geq \frac{\lambda_r^{k-1}}{(k-1)!}. \quad (3.3)$$

This one-large-action contribution will determine the sharp exponential scale. The case $r = 1$ must be treated separately: then there is only one multiplicity vector, not k distinct placements of a distinguished action.

3.2 Finite truncations

Let $\hat{U}_N$ solve

$$\hat{U}_N = \sum_{j=1}^N q^j e^{\lambda_j \hat{U}_N}. \quad (3.4)$$

Its defining function is analytic near $(0, 0)$, and its derivative with respect to the left-hand unknown at the origin is one. Thus $\hat{U}_N$ is a convergent germ. The formal locality above gives

$$[q^n] \hat{U}_N = [q^n] \hat{U} \quad (n \leq N). \quad (3.5)$$

It is precisely the lack of a uniform analytic neighborhood as N grows that allows the infinite formal limit to diverge.

4 The exact convergence criterion and positive certificates

Proposition 4.1 (Convergence at a positive point). *For $0 < q < 1$, the nonnegative series $\sum_n u_n q^n$ is finite if and only if there exists $r > 0$ such that*

$$\sum_{j \geq 1} q^j e^{\lambda_j r} \leq r. \quad (4.1)$$

When finite, its sum is the least nonnegative fixed point of $r \mapsto \Phi_\lambda(q, r)$.

Proof. Assume (4.1). Monotone real iteration from zero remains in $[0, r]$. Its coefficients are those of the formal iterates in Lemma 3.1, are nonnegative, and increase to u_n . The monotone convergence theorem, applied to these nonnegative series, gives $\sum_n u_n q^n \leq r$. Applying the same theorem to the exponential expansions verifies the fixed-point identity. Iteration from zero is below every nonnegative fixed point, proving minimality.

Conversely, if $u = \sum_n u_n q^n < \infty$, substitute the sum into the formal identity and use Tonelli's theorem for the nonnegative terms. It gives $u = \sum_j q^j e^{\lambda_j u}$. Since $u \geq q > 0$, the value $r = u$ is an admissible supersolution. $\square$

Proof of the convergence part of Theorem 2.2. If $\lambda_j \leq Mj$ for all j , then for sufficiently small $|q|$ and bounded $|u|$,

$$\sum_j |q|^j e^{\lambda_j |u|} \leq \sum_j (|q| e^{M|u|})^j.$$

Hence Φ_λ is holomorphic in a product neighborhood of $(0, 0)$. The analytic implicit-function theorem applies to $u - \Phi_\lambda(q, u)$ because its u -derivative at the origin is one.

If $\sup_j \lambda_j/j = \infty$, take any $0 < q < 1$ and $r > 0$. Along a subsequence,

$$j \log q + \lambda_j r = j \left(\log q + r \frac{\lambda_j}{j} \right) \longrightarrow +\infty.$$

The summands in (4.1) do not even tend to zero. Proposition 4.1 excludes convergence at every positive q . Nonnegativity of the coefficients then implies radius zero. $\square$

Remark 4.2. The divergence proof is not merely an upper-majorant failure. Positivity makes the supersolution criterion necessary as well as sufficient. For signed or complex coefficients this equivalence generally disappears.

4.1 Quantitative certificates inside a convergence domain

Suppose, for $Q, r > 0$, that

$$\Phi_\lambda(Q, r) \leq r, \quad \kappa = \partial_u \Phi_\lambda(Q, r) = \sum_j Q^j \lambda_j e^{\lambda_j r} < 1. \quad (4.2)$$

Then the fixed-point map is a contraction on $|u| \leq r$ for $|q| \leq Q$. The infinite solution and the solution U_N of the first N actions obey

$$|U(q) - U_N(q)| \leq \frac{\sum_{j > N} Q^j e^{\lambda_j r}}{1 - \kappa}, \quad |q| \leq Q. \quad (4.3)$$

This follows by subtracting the two fixed-point equations and bounding the derivative on the line segment joining their values.

A different certificate bounds truncation by total action rather than by primitive action. If merely $\Phi_\lambda(Q, r) \leq r$, then for $|q| < Q$,

$$\left| U(q) - \sum_{n=1}^N u_n q^n \right| \leq r \left(\frac{|q|}{Q} \right)^{N+1}. \quad (4.4)$$

Indeed, factor $(|q|/Q)^{N+1}$ from the nonnegative coefficient tail and use $\sum_n u_n Q^n \leq r$. Equations (4.3) and (4.4) certify different truncation operations; they should not be interchanged without checking their hypotheses.

5 A coefficient-envelope lemma

This section contains the quantitative step behind both the asymptotic formula and the exact Gevrey type. The details matter: replacing the number of compositions by 2^n throughout would lose a constant of order n and would not prove (2.2).

Fix $p > 1$, $\beta \in \mathbb{R}$, and write

$$s = p - 1, \quad f(r) = r^p (\log(r+1))^\beta.$$

For $A > 0$ define

$$X_n(A) = sn \log n + (\beta - p)n \log \log n + \left\lceil \log A + p \log \frac{p}{s} - s \right\rceil n. \quad (5.1)$$

Lemma 5.1 (Sharp logarithmic envelope). *For every $A > 0$ and $D \geq 0$,*

$$\max_{2 \leq r \leq n} \log \frac{(Af(r))^{n-r}}{(n-r)!} = X_n(A) + o(n), \quad (5.2)$$

$$\log \sum_{r=1}^n \binom{n-1}{r-1} \frac{(Af(r) + D(n-r+1))^{n-r}}{(n-r+1)!} = X_n(A) + o(n). \quad (5.3)$$

The relevant values of r satisfy $r \log n/n \rightarrow p/s$ at the exponential scale. More precisely, restricting the upper sum to $|r \log n/n - p/s| \geq \varepsilon$ lowers its logarithm by $c_\varepsilon n + o(n)$ for some $c_\varepsilon > 0$.

Proof. Put $L = \log n$, $k = n - r + 1$, and $M = Af(r) + Dk$. The logarithm of an upper-sum term is

$$E_n(r) = \log \binom{n-1}{r-1} + (k-1) \log M - \log k!. \quad (5.4)$$

All occurrences of M have polynomial growth in n times a power of $\log n$, and M is bounded away from zero. Uniform Stirling estimates therefore permit errors $O(\log n)$ when replacing factorials by their leading logarithms.

A lower comparison. Choose

$$r = \left\lfloor \frac{p n}{s L} \right\rfloor.$$

Then $r \rightarrow \infty$, $r/n \rightarrow 0$, and

$$\log f(r) = pL + (\beta - p) \log L + p \log(p/s) + o(1).$$

Substituting this into the lower expression and using Stirling gives $X_n(A) + o(n)$. Since the corresponding upper term differs from this lower term by the factor $\binom{n-1}{r-1}/k$ and by replacing $Af(r)$ with a larger quantity, its logarithm is also at least $X_n(A) + o(n)$. Here $\log k = o(n)$.

Localization of any maximizing row. For $r \geq \delta n$ with fixed $\delta > 0$, we have $k \leq (1 - \delta)n + 1$ and

$$E_n(r) \leq (1 - \delta)snL + O(n \log L) + O(n).$$

The entropy bound $\log \binom{n-1}{r-1} \leq n \log 2$ is sufficient *at this preliminary step*. Such rows are smaller than the lower comparison by a positive multiple of nL . A maximizing row thus has $r/n \rightarrow 0$. For these rows

$$\log \binom{n-1}{r-1} = o(n). \quad (5.5)$$

Write $\delta_n = r/n$. Comparing with the lower value now forces $\log M \geq pL - O(\log L)$. Consequently $M/n \rightarrow \infty$, the term Dk is negligible in M , and

$$\log r = L - O(\log L), \quad \log \log(r+1) = \log L + o(1).$$

At such a row, Stirling and (5.5) give

$$\frac{E_n(r)}{n} = (1 - \delta_n)[sL + p \log \delta_n + \beta \log L + \log A - \log(1 - \delta_n) + 1] + o(1).$$

The same comparison, using $\log \delta_n \leq 0$, implies $\delta_n L = O(\log L)$. Hence $\delta_n \log L = o(1)$ and $\delta_n \log \delta_n = o(1)$. The previous display reduces to

$$\frac{E_n(r)}{n} = sL + (\beta - p) \log L + \log A + 1 + p \log y - sy + o(1), \quad y = \frac{rL}{n}. \quad (5.6)$$

The one-dimensional maximum. The function

$$g(y) = p \log y - sy \quad (y > 0)$$

is strictly concave, tends to $-\infty$ at both ends, and has its unique maximum at $y_* = p/s$. Its maximum value is $p \log(p/s) - p$. Substitution into (5.6) proves that the largest upper term has logarithm at most $X_n(A) + o(n)$. A sum of n terms changes the logarithm by at most $\log n = o(n)$, proving (5.3). The upper expression also bounds every lower expression up to the harmless factor k , because the binomial factor is at least one. This proves (5.2).

For the restricted sum, a contradiction argument gives the same localization for any sequence of rows whose values are within $O(n)$ of the maximum. On the closed set $|y - y_*| \geq \varepsilon$, strict concavity and the behavior at the ends give $g(y) \leq g(y_*) - c_\varepsilon$. Rows outside the localized range have already lost more than a fixed multiple of n . Applying (5.6) proves the final assertion. This argument also makes the $o(n)$ estimates uniform over the rows needed for that assertion. $\square$

5.1 From eventual convexity to a uniform upper bound

Lemma 5.2 (Convex majorant). *Suppose $\lambda_j \sim af(j)$. For every $A > a$, there is $D \geq 0$ such that every composition $j_1 + \dots + j_k = n$, $j_i \geq 1$, satisfies*

$$\lambda_{j_1} + \dots + \lambda_{j_k} \leq Af(n - k + 1) + Dk. \quad (5.7)$$

Proof. Direct differentiation gives $f''(t) \sim p(p-1)t^{p-2}(\log t)^\beta > 0$. Thus f is convex and increasing eventually. Extend its tail to a convex function h on $[1, \infty)$ by continuing a tangent line to the left. Then $h = f$ on a tail. Since $A > a$, increasing a constant D_0 handles the finitely many exceptional indices and gives $\lambda_j \leq Ah(j) + D_0$ for all j .

For a convex function, shifting mass from the smaller of two arguments larger than one to the larger cannot decrease their sum. Repeating gives

$$\sum_{i=1}^k h(j_i) \leq h(n - k + 1) + (k - 1)h(1).$$

On the finite interval where $h \neq f$, their difference is bounded. Absorbing this bound and $Ah(1) + D_0$ into a nonnegative D proves the claim. This also explains why arbitrarily large but finitely many initial slopes do not alter the asymptotic constant. $\square$

6 Logarithmic asymptotics and concentration

Proof of the coefficient part of Theorem 2.3. For any $A > a$, apply Lemma 5.2 in (3.2). There are $\binom{n-1}{k-1} = \binom{n-1}{r-1}$ compositions with $k = n - r + 1$ parts. Lemma 5.1 therefore gives

$$\log u_n \leq X_n(A) + o(n).$$

For a lower bound, choose $r = \lfloor (p/s)n / \log n \rfloor$ in (3.3). Since $\lambda_r / (af(r)) \rightarrow 1$, the lower calculation in Lemma 5.1 gives

$$\log u_n \geq X_n(a) + o(n).$$

Letting $A \downarrow a$ proves (2.4). $\square$

6.1 Concentration of the total excess action

The integer $n - K = \sum_j (j - 1)m_j$ is the total action remaining after giving one unit to each of the K factors. It is a more useful variable than K itself: almost all factors have unit action on the leading scale.

Proposition 6.1. *Under the assumptions of Theorem 2.3, for every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that*

$$\Pr\left(\left|\frac{(n - K + 1) \log n}{n} - \frac{p}{p - 1}\right| \geq \varepsilon\right) \leq \exp(-c_\varepsilon n + o(n)).$$

Proof. The total unnormalized weight at a fixed value of $K = k$ is exactly the k -row of (3.2). Bound the rows by Lemma 5.2. The restricted-envelope conclusion in Lemma 5.1 gives a strict exponential loss outside the stated window. Choose $A > a$ sufficiently close to a that the increase $n \log(A/a)$ is smaller than half that loss. Divide by the lower estimate $u_n = \exp(X_n(a) + o(n))$. $\square$

6.2 Concentration on one large action

Proposition 6.2. *Under the same assumptions, for every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that*

$$\Pr(J_{\max} \leq (1 - \varepsilon)(n - K + 1)) \leq \exp(-c_\varepsilon n + o(n)).$$

Proof. It suffices to consider $0 < \varepsilon < 1/2$, since increasing ε makes the event smaller. By Proposition 6.1, we may restrict to $r = n - k + 1$ in a fixed compact multiple of $n / \log n$, losing only an exponentially small probability. In this range $k \sim n$ and $f(r)/n \rightarrow \infty$.

Set $b = \lfloor (1 - \varepsilon)r \rfloor$. For a convex function h as in Lemma 5.2, maximize $\sum_i h(j_i)$ subject to $1 \leq j_i \leq b$ and $\sum_i (j_i - 1) = r - 1$. Since $b > r/2$ for large r , a maximizing configuration has at most two entries greater than one: one at b , the other at $r - b + 1$. This follows by the same pairwise mass-transfer argument, stopping when an entry reaches the cap. Thus

$$\sum_i \lambda_{j_i} \leq A\{f(b) + f(r - b + 1)\} + D'k.$$

Uniformly in the restricted range of r ,

$$\frac{f(b) + f(r - b + 1)}{f(r)} \rightarrow d_\varepsilon := (1 - \varepsilon)^p + \varepsilon^p < 1.$$

The $D'k$ term is negligible. Choose $A > a$ close enough that $(A/a)d_\varepsilon < 1$. Compared with the unrestricted row bound, the capped slope sum has a fixed multiplicative loss. Its power is $k - 1 \sim n$, so the row loses $cn + o(n)$ in its logarithm. The composition-count entropy is $o(n)$ in this range and cannot repair the loss. Summing the at most n rows and dividing by $u_n = \exp(X_n(a) + o(n))$ completes the proof. $\square$

Together these propositions finish Theorem 2.3. They justify the one-large-action mechanism at exponential precision. They do *not* give a local limit theorem, a fluctuation scale about the true finite- n center, or a full multiplicative asymptotic for u_n .

Editorial note (ProveIt, 2026-09-29). For power slopes these gaps are closed by later packages in this series; see the editorial note after the question “Beyond logarithmic coefficient asymptotics” in Section 13.

7 The exact Gevrey-type identity

Proof of (2.2). Fix $s > 0$ and put $p = s + 1$. Denote $A_s(\boldsymbol{\lambda})$ by A_* . First suppose $A_* < \infty$. For every $A > A_*$ there is $D \geq 0$ such that

$$\lambda_j \leq \tilde{\lambda}_j := A(j \log(j+1))^p + D$$

for every j . Coefficientwise monotonicity gives $u_n(\boldsymbol{\lambda}) \leq u_n(\tilde{\boldsymbol{\lambda}})$. Apply Theorem 2.3 to $\tilde{\lambda}_j$ with $\beta = p$:

$$\log u_n(\tilde{\boldsymbol{\lambda}}) = sn \log n + [\log A + p \log(p/s) - s]n + o(n).$$

Since $\log n! = n \log n - n + o(n)$,

$$T_s(\hat{U}) \leq A(p/s)^p.$$

Let $A \downarrow A_*$. This also proves the upper bound when $A_* = 0$.

For the lower bound, take any $b > 0$ with $b < A_*$; when $A_* = \infty$, b is arbitrary. Infinitely many j satisfy

$$\lambda_j \geq b(j \log j)^p.$$

Along these indices, choose

$$k = \left\lfloor \frac{s}{p} j \log j \right\rfloor, \quad n = j + k.$$

The multiplicities $m_j = 1$ and $m_1 = k$ give $u_n \geq \lambda_j^k / k!$. To record the sharp constant explicitly, put $c = s/p$ and $L = \log j$. Then

$$k = cjL + O(1), \quad n = cjL + j + O(1), \quad \log n = L + \log L + \log c + o(1).$$

Stirling’s formula, with the relative difference j/n retained, gives

$$\begin{aligned} \frac{1}{n} \log \frac{\lambda_j^k}{k!(n!)^s} &\geq \log b - p \log c + p - \frac{s}{c} + o(1) \\ &= \log b + p \log(p/s) + o(1). \end{aligned} \tag{7.1}$$

Therefore $T_s(\hat{U}) \geq b(p/s)^p$. Letting $b \uparrow A_*$ proves the lower bound, including infinity. When $A_* = 0$, the lower bound is simply nonnegativity. This proves the identity in every case. $\square$

7.1 Consequences and endpoint behavior

Corollary 7.1 (Borel radii and zero type). *For $s > 0$, the factorial-normalized transform $\sum_n u_n t^n / (n!)^s$ has radius*

$$R_s = \frac{1}{C_s A_s(\boldsymbol{\lambda})}, \quad (7.2)$$

with $1/0 = \infty$ and $1/\infty = 0$. In particular it is entire exactly when $\lambda_j = o((j \log j)^{s+1})$. For the normalization $\Gamma(1 + sn)$, the radius is instead $s^s / (C_s A_s(\boldsymbol{\lambda}))$.

Proof. The first statement is Cauchy–Hadamard and Theorem 2.2. Stirling gives $\Gamma(1 + sn)^{1/n} / (n!)^{s/n} \rightarrow s^s$, which proves the last assertion. $\square$

Corollary 7.2 (The least Gevrey index). *Let*

$$\rho = \limsup_{j \rightarrow \infty} \frac{\log(1 + \lambda_j)}{\log j}.$$

Then

$$\inf\{s \geq 0 : \widehat{U} \in \mathcal{G}_s\} = \max(0, \rho - 1), \quad (7.3)$$

with value infinity when $\rho = \infty$. Membership at the infimum is not automatic.

Proof. For $s > \max(0, \rho - 1)$, a positive power of j separates λ_j from $(j \log j)^{s+1}$ eventually, so (2.3) applies. If $0 < s < \rho - 1$, the reverse separation holds on a subsequence and excludes $\mathcal{G}_s$. The case $s = 0$ is covered by the convergence criterion. These implications prove the infimum, without making an unwarranted assertion about its attainment. $\square$

For $\lambda_j \sim a j^p (\log j)^\beta$ with $p > 1$, the critical index is $s = p - 1$. At that index the type is zero when $\beta < p$, is $a(p/(p-1))^p$ when $\beta = p$, and is infinite when $\beta > p$. Thus the logarithmic factor decides whether the critical Gevrey class is attained, and whether its Borel radius is infinite, finite, or zero.

Example 7.3 (Sparse spikes are detected sharply). Fix $s > 0$ and $a > 0$, and let

$$\lambda_j = \begin{cases} a(j \log j)^{s+1}, & j = 2^{2^m}, m \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then $T_s(\widehat{U}) = aC_s$. The density of the large slopes is irrelevant to this limsup statement. We do not infer a full coefficient asymptotic from this sparse example; the regular-variation hypothesis of Theorem 2.3 does not hold.

8 Borel growth and the positive-direction obstruction

8.1 An elementary entire-function lemma

Lemma 8.1. *Let $b_n \geq 0$ for all n , with $b_n > 0$ eventually, and suppose, for $p > 0$ and $D \in \mathbb{R}$, that*

$$\log b_n = -pn \log \log n + Dn + o(n). \quad (8.1)$$

Then $B(t) = \sum_{n \geq 1} b_n t^n$ is entire and

$$\lim_{t \rightarrow +\infty} \frac{\log \log B(t)}{t^{1/p}} = e^{D/p}. \quad (8.2)$$

The same conclusion holds after changing finitely many coefficients to arbitrary nonnegative values.

Proof. Equation (8.1) gives $b_n^{1/n} \rightarrow 0$, so B is entire. Put $c = e^{D/p}$. For any $0 < c_- < c < c_+$, eventually

$$\left(\frac{c_-}{\log n}\right)^{pn} \leq b_n \leq \left(\frac{c_+}{\log n}\right)^{pn}.$$

Write $T = t^{1/p}$. For a lower bound, fix $0 < d < c_-$ and select $n = \lfloor e^{dT} \rfloor$. The logarithm of this single term satisfies

$$\log(b_n t^n) \geq pn \log(c_-/d) + o(n).$$

It follows that $\liminf \log \log B(t)/T \geq d$. Let $d \uparrow c$ through admissible choices of c_- .

For an upper bound choose $d > c_+$ and $N = \lceil e^{dT} \rceil$. When $n > N$, the upper bound gives $b_n t^n \leq (c_+/d)^{pn}$, up to an immaterial adjustment at the cutoff. The tail is bounded by a convergent geometric series. Since the entire coefficient sequence has bounded n th roots, there is $K \geq 1$ such that $b_n \leq K^n$ for all n . Therefore the first N terms are bounded by $N \max(1, Kt)^N$. Taking two logarithms yields

$$\log \log B(t) \leq dT + O(\log \log(t + e)).$$

Let $d \downarrow c$ through choices of c_+ . Finite initial terms do not change either estimate. $\square$

Proof of Theorem 2.4. Take $\beta = 0$ in Theorem 2.3 and put $s = p - 1$. After subtracting $s \log n!$, it gives

$$\log \frac{u_n}{(n!)^s} = -pn \log \log n + \left[\log a + p \log \frac{p}{p-1} \right] n + o(n).$$

Lemma 8.1 gives (2.6). For $p = 2$ and $a = 1$, the resulting growth is faster than $\exp(Kt)$ for every fixed K . In fact $\log B_1(t) = \exp((2 + o(1))\sqrt{t})$, which dominates t/q for every $q > 0$. Thus the positive integrand in

$$\frac{1}{q} \int_0^\infty e^{-t/q} B_1(t) dt$$

eventually tends to infinity, so the integral diverges. $\square$

8.2 The standard Gamma normalization

For general $s = p - 1$, another common normalization is

$$\tilde{B}_s(t) = \sum_{n \geq 1} \frac{u_n}{\Gamma(1 + sn)} t^n.$$

Stirling changes the constant D in Lemma 8.1 by $-s \log s$. Consequently

$$\lim_{t \rightarrow +\infty} \frac{\log \log \tilde{B}_s(t)}{t^{1/p}} = a^{1/p} \frac{p}{s} s^{-s/p}. \quad (8.3)$$

It follows that $\tilde{B}_s$ grows faster on the positive ray than $\exp(Kt^d)$ for every finite $d, K > 0$. In particular it fails the growth requirement for the usual order- $1/s$ Borel–Laplace summation procedure. This is a directional conclusion. No corresponding assertion about the negative ray is proved here.

8.3 Why positivity makes the obstruction especially transparent

For the quadratic case, Tonelli gives a second proof of the failure: if the displayed positive-ray Laplace integral were finite, then

$$\frac{1}{q} \int_0^\infty e^{-t/q} \sum_{n \geq 1} \frac{u_n t^n}{n!} dt = \sum_{n \geq 1} u_n q^n$$

would be finite. The convergence criterion says the right-hand side is infinite. The explicit double-logarithmic asymptotic is stronger: it identifies how the growth condition fails rather than merely proving a contradiction.

Remark 8.2 (What “entire” does not settle). The Borel transform here has no finite-plane singularities, but has very large growth at infinity. We therefore do not infer ordinary summability from endless analytic continuation alone, nor do we assign finite Borel singularities or Stokes constants without an additional argument. The separation between analytic continuation and Laplace growth is essential in the summability framework [7].

9 Sectorial analytic realization without slope bounds

The formal series may have zero convergence radius and may fail every finite Gevrey bound. Nevertheless its defining equation has a literal analytic realization in suitable directions. The reason is a different estimate: in the left half-plane, $e^{\lambda_j u}$ suppresses rather than amplifies large positive slopes.

Theorem 9.1 (Quantitative left-sector construction). *Let $\lambda_j \geq 0$ be arbitrary and fix $\eta \in (0, 1)$. Set*

$$K = 2e\lambda_1 + 2.$$

Choose $r_0 > 0$ sufficiently small that

$$r_0 \leq \frac{1}{4}, \quad Kr_0 \leq \min(1, \eta/2), \quad 2\lambda_1 r_0 \leq 1, \quad (9.1)$$

and

$$\kappa_0 := r_0 \left(\lambda_1 + \frac{2}{e\eta(1-r_0)} \right) \leq \frac{1}{2}. \quad (9.2)$$

For every $q \in S_{\eta, r_0}$, the map $u \mapsto \Phi_{\lambda}(q, u)$ is a contraction of the closed disk

$$D_q = \{u : |u - q| \leq K|q|^2\}$$

into itself. Its fixed point $U(q)$ is holomorphic in q , satisfies $U(q) = q + O(q^2)$, and is the unique solution with this latter behavior on sufficiently small proper subsectors.

Let $U_N(q)$ be the fixed point in the same disk for the first N actions. Then, for $r = |q|$,

$$\boxed{|U(q) - U_N(q)| \leq \frac{r^{N+1}}{(1-r)(1-\kappa_0)}}. \quad (9.3)$$

For each fixed N , U_N agrees near the origin in the sector with the analytic germ defined by (3.4). The function U has Poincaré expansion $\widehat{U}$.

Proof. Write $r = |q|$. If $u \in D_q$, the choices above imply

$$|u| \leq 2r, \quad \Re u \leq -\eta r/2 < 0.$$

The infinite sum defining Φ_{λ} is locally normally convergent on $|q| < 1$, $\Re u < 0$, since $|q^j e^{\lambda_j u}| \leq |q|^j$. It is therefore jointly holomorphic there.

Separate the first action from the rest. The bound $|e^z - 1| \leq |z|e^{|z|}$ gives

$$\begin{aligned} |\Phi_{\lambda}(q, u) - q| &\leq r|e^{\lambda_1 u} - 1| + \sum_{j \geq 2} r^j \\ &\leq 2e\lambda_1 r^2 + \frac{r^2}{1-r} \leq Kr^2. \end{aligned}$$

Thus the disk is invariant. For $a > 0$ and $t \geq 0$, $te^{-at} \leq 1/(ea)$. Using this with $a = \eta r/2$ gives

$$\begin{aligned} |\partial_u \Phi_\lambda(q, u)| &\leq \lambda_1 r + \sum_{j \geq 2} r^j \lambda_j e^{-\lambda_j \eta r/2} \\ &\leq \lambda_1 r + \frac{2}{e\eta r} \frac{r^2}{1-r} \leq \kappa_0. \end{aligned}$$

The same estimates hold after truncating the sum at any $N \geq 1$. Banach's contraction theorem supplies the fixed points. Picard iterates starting at q are holomorphic on the sector and converge uniformly on compact subsets, so their limit is holomorphic.

Any other solution $V = q + O(q^2)$ in a proper left subsector has $|V| \leq 2r$ and $\Re V \leq -\eta r/2$ after reducing r_0 if necessary. The invariance estimate then places $V = \Phi_\lambda(q, V)$ in D_q , where uniqueness has already been proved.

Subtract the infinite and finite fixed-point equations. The segment joining U and U_N lies in D_q , so

$$|U - U_N| \leq \kappa_0 |U - U_N| + \sum_{j > N} r^j e^{\lambda_j \Re U} \leq \kappa_0 |U - U_N| + \frac{r^{N+1}}{1-r}.$$

This proves (9.3).

For a fixed N , the analytic implicit-function theorem at $(0, 0)$ gives the finite-action germ. It has expansion $q + O(q^2)$ and hence coincides with the disk fixed point near the origin in the sector. By (3.5), its first N coefficients are $u_1, \dots, u_N$. For that fixed N ,

$$U_N(q) = \sum_{n=1}^N u_n q^n + O_N(q^{N+1}).$$

Combining this with (9.3) proves the Poincaré assertion. The constants hidden in O_N may depend on N ; no uniform Gevrey remainder estimate has been used or claimed. $\square$

9.1 The direction of approach is substantive

If $\sup_j \lambda_j/j = \infty$, no branch satisfying $U(q)/q \rightarrow 1$ as $q \downarrow 0$ on the positive real axis can solve the literal identity. Indeed, then $\Re U(q) > 0$ for sufficiently small positive q , and the terms $q^j e^{\lambda_j U(q)}$ fail to tend to zero along a subsequence. By contrast, Theorem 9.1 gives a real negative solution when q is real negative, because the equation and the invariant disk are preserved by complex conjugation.

Under $q = e^{-x}$, negative q corresponds to a complex horizontal line with $\Im x$ an odd multiple of π , rather than to real $x \rightarrow +\infty$. Thus the left-sector construction is a complex realization of the same formal exponential transseries, not a positive-real realization hidden by a change of notation.

9.2 A practical implication for summation algorithms

The sequence $U_N(q)$, obtained by retaining primitive actions and solving a finite implicit equation, converges geometrically on these sectors with the explicit certificate (9.3). The Taylor partial sums $\sum_{n \leq N} u_n q^n$ may diverge at every nonzero q . These are different approximation schemes. An implementation should record which scheme it uses, rather than labeling both merely as “truncation of the transseries.”

10 Transfer to genuine compositional inversion

Because $u_1 = 1$, the formal solution has a unique inverse $\widehat{Q} \in q + q^2\mathbb{R}[[q]]$ under composition. In particular,

$$\widehat{Q}(\widehat{U}(q)) = q, \quad u = \sum_{j \geq 1} \widehat{Q}(u)^j e^{\lambda_j u} \quad (10.1)$$

as formal identities. The following classical stability fact is proved here to keep the application independent of analytic realization choices.

Proposition 10.1 (Gevrey stability under formal reversion). *If $\widehat{f}(q) = q + \sum_{n \geq 2} f_n q^n$ and $s \geq 0$, then*

$$\widehat{f} \in \mathcal{G}_s \iff \widehat{f}^{(-1)} \in \mathcal{G}_s.$$

Proof. For $s = 0$ this is the ordinary analytic inverse-function theorem in both directions. We give a coefficient proof valid also for $s > 0$. Assume $|f_n| \leq CA^n(n!)^s$, increasing A, C to at least one. Write $\widehat{f} = q(1 + h(q))$, where $h(q) = \sum_{m \geq 1} f_{m+1}q^m$. Lagrange inversion gives, for $n \geq 2$,

$$\begin{aligned} [q^n]\widehat{f}^{(-1)} &= \frac{1}{n}[z^{n-1}](1 + h(z))^{-n} \\ &= \frac{1}{n} \sum_{k=1}^{n-1} (-1)^k \binom{n+k-1}{k} \sum_{\substack{m_1 + \dots + m_k = n-1 \\ m_i \geq 1}} \prod_i f_{m_i+1}. \end{aligned}$$

For positive integers m_i with sum $n - 1$,

$$\prod_{i=1}^k (m_i + 1)! \leq 2^{k-1} (n - k + 1)! \leq 2^n n!. \quad (10.2)$$

One way to see the first inequality is to transfer a unit from a smaller factorial argument greater than two to a larger one; this cannot decrease the product, and it leaves all but one argument equal to two.

The binomial coefficient in the reversion formula is at most 4^n , the number of compositions is at most 2^n , and $\prod_i CA^{m_i+1} \leq C^n A^{2n}$. Equation (10.2) therefore bounds the inverse coefficient by $B^n(n!)^s$ for a fixed $B > 0$, after absorbing all exponential factors. This proves one implication. Apply it again to the inverse series to obtain the other. $\square$

Corollary 10.2 (The classification survives inversion). *The inverse $\widehat{Q}$ has positive convergence radius exactly when $\lambda_j = O(j)$. For each $s > 0$,*

$$\widehat{Q} \in \mathcal{G}_s \iff \lambda_j = O((j \log(j+1))^{s+1}).$$

Its least Gevrey index is the value in (7.3).

Remark 10.3. This corollary transfers *class membership*, not the numerical type T_s . The inverse coefficients need not be nonnegative. We have not proved that $T_s(\widehat{Q}) = T_s(\widehat{U})$, nor that the concentration theorem transfers through the cancellations in reversion.

11 Examples, benchmarks, and sharp boundaries

11.1 Analytic benchmark kernels

When all slopes vanish, $U = q/(1 - q)$ has radius one. If $\lambda_j = 1$ for every j , then

$$U = \frac{q}{1 - q} e^U, \quad U = -W_0\left(-\frac{q}{1 - q}\right).$$

Its positive critical point is $U = 1$, $q = 1/(1 + e)$; the positive supersolution criterion and the nonzero quadratic derivative at the critical point show that this is its radius of convergence.

For $\lambda_j = j$, the kernel is geometric in qe^U , and

$$U = \frac{qe^U}{1 - qe^U} \iff U = q(1 + U)e^U. \quad (11.1)$$

Writing $q = Ue^{-U}/(1 + U)$, the positive maximum occurs at

$$\tau = \frac{\sqrt{5} - 1}{2}, \quad R = \frac{\tau e^{-\tau}}{1 + \tau} = \tau^2 e^{-\tau}.$$

The branch at the origin reaches this simple critical point. Local analytic inversion gives

$$U(q) = \tau - \sqrt{\frac{2(1 + \tau)}{3 + \tau}} \sqrt{1 - q/R} + O(1 - q/R). \quad (11.2)$$

This is a familiar analytic implicit-function singularity; the new phenomena begin when the countable slopes outgrow their actions.

11.2 Quadratic feedback

For $\lambda_j = j^2$,

$$\begin{aligned} \widehat{U}(q) = & q + 2q^2 + \frac{15}{2}q^3 + \frac{107}{3}q^4 + \frac{1555}{8}q^5 + \frac{17362}{15}q^6 \\ & + \frac{5288311}{720}q^7 + \frac{15414767}{315}q^8 + \dots \end{aligned}$$

It has radius zero, is Gevrey-1 of type zero, and is not Gevrey- s for any $s < 1$. Its coefficients satisfy

$$\log u_n = n \log n - 2n \log \log n + (2 \log 2 - 1)n + o(n). \quad (11.3)$$

Thus “Gevrey-1 of type zero” is compatible with divergence: the normalization by $n!$ is stronger than the normalization by an exponential. The ordinary Borel transform is entire but has the positive-ray growth proved in Section 8.

11.3 The critical logarithmic correction

For $\lambda_j = a(j \log(j + 1))^2$, the exact type identity gives

$$T_1(\widehat{U}) = 4a, \quad R_{\text{Borel}} = \frac{1}{4a}.$$

This finite Borel radius is obtained without a full multiplicative asymptotic for the coefficients. Increasing the logarithmic exponent to $2 + \varepsilon$ destroys Gevrey-1 membership; decreasing it below two makes the ordinary Borel transform entire.

Slope sequence	Convergence	Gevrey conclusion
$\lambda_j = O(j)$	Positive radius	$\mathcal{G}_0$
$j \log(j + 1)$	Radius zero	Every $\mathcal{G}_s$, $s > 0$; not $\mathcal{G}_0$
j^2	Radius zero	Critical class $\mathcal{G}_1$, type zero
$a(j \log(j + 1))^2$	Radius zero	Critical class $\mathcal{G}_1$, type $4a$
$j^2(\log(j + 1))^3$	Radius zero	Every $\mathcal{G}_s$, $s > 1$; not $\mathcal{G}_1$
j^3	Radius zero	Critical class $\mathcal{G}_2$, type zero
$e^{\sqrt{j}}$	Radius zero	No finite Gevrey class

All rows nevertheless admit the left-sector realization of Theorem 9.1. The last row is a particularly strong warning against inferring coefficient regularity from sectorial existence alone.

11.4 Why a complete black-box decision procedure is impossible

The sharp criteria are mathematical equivalences, not a claim that their asymptotic conditions can be decided from an arbitrary program for λ_j .

Proposition 11.1. *There is no algorithm that takes an arbitrary program computing a nonnegative integer slope sequence and always decides whether its formal solution is convergent. For any fixed $s > 0$, there is likewise no such algorithm deciding membership in $\mathcal{G}_s$.*

Proof. Given a machine M , define a total computable sequence by running M for j steps and setting

$$\lambda_j = \begin{cases} j^2, & M \text{ has not halted within those steps,} \\ 0, & M \text{ has halted within those steps.} \end{cases}$$

If M halts, only finitely many slopes are nonzero and the solution is convergent. If M never halts, the sequence is j^2 and the solution is divergent. A proposed decision procedure would decide the halting problem. For a fixed positive s , replace j^2 by j^m for an integer $m > s + 1$. The nonhalting case is not in $\mathcal{G}_s$, while the halting case remains convergent. The same reduction applies. $\square$

The appropriate implementation goal is consequently a sound certificate system for structured input families, not an unconditional classifier for arbitrary programs. This reduction is a standard use of undecidability; it is not presented as a new theorem of computability theory.

12 Reproducible coefficient checks and numerical diagnostics

The supplementary script `code/verify.py` uses exact integers for coefficient generation when the slopes are nonnegative integers. It also uses an independent partition enumeration to check the Lagrange formula. No numerical experiment is used as a substitute for a proof above.

12.1 An exact recurrence

Put

$$A_n = n!u_n, \quad E_{j,k} = k![q^k]e^{\lambda_j \widehat{U}(q)}, \quad E_{j,0} = 1.$$

Differentiating the exponential formally gives

$$E_{j,k} = \lambda_j \sum_{r=1}^k \binom{k-1}{r-1} A_r E_{j,k-r}. \quad (12.1)$$

The original fixed-point identity gives

$$A_n = \sum_{j=1}^n \frac{n!}{(n-j)!} E_{j,n-j}. \quad (12.2)$$

All quantities needed on the right have smaller index than n , except for the already known constant $E_{n,0} = 1$. Integer slopes therefore produce integer A_n , even when u_n is not integral. The supplied implementation takes $O(N^3)$ integer arithmetic operations and $O(N^2)$ stored integers through order N ; this is an arithmetic-operation count, not a bit-complexity bound.

The executed checks were as follows. For $\lambda_j = j^2$, coefficients through order 120 were computed exactly. The first twelve were checked against direct partition enumeration. Twelve zero-slope coefficients were checked against $q/(1-q)$, and twelve linear-slope coefficients against an independent Lagrange formula for (11.1). All thirty-six comparisons passed. The exact values and the verification record are included in `data/`.

12.2 A rigorous sandwich, evaluated approximately

For the quadratic model and $k = n - r + 1$, convexity gives the exact maximum

$$\sum_{i=1}^k j_i^2 \leq r^2 + k - 1.$$

Consequently

$$u_n \geq \max_{2 \leq r \leq n} \frac{(r^2 + n - r)^{n-r}}{(n - r)!}, \quad (12.3)$$

$$u_n \leq \sum_{r=1}^n \binom{n-1}{r-1} \frac{(r^2 + n - r)^{n-r}}{(n - r + 1)!}. \quad (12.4)$$

For numerical work the all-unit contribution can also be included in the lower maximum, with its correct denominator $n!$. The mathematical inequalities are rigorous. The decimal evaluations below use ordinary floating-point logarithms, not directed rounding, and are diagnostics rather than interval certificates.

Define the normalized logarithm

$$H_n = \frac{\log u_n}{n} - \log n + 2 \log \log n.$$

Theorem 2.3 proves $H_n \rightarrow 2 \log 2 - 1 \approx 0.38629436$. Applying the normalization to the lower and upper bounds gives:

n	Lower value	Upper value	Lower-bound maximizing r	$r \log n / n$
100	1.017199	1.626109	38	1.749965
1 000	1.059262	1.666026	292	2.017065
10 000	1.031046	1.587396	2 332	2.147851
100 000	0.990011	1.493908	19 183	2.208524
1 000 000	0.949010	1.406654	161 926	2.237090

These values make an important caution visible: convergence of the normalized logarithm and of the leading saddle scale is slow. A moderate coefficient table would not reliably identify the limiting constant or justify a full coefficient asymptotic. The table is not evidence that the finite- n maximizing action is already close to its limiting value two.

12.3 Reproduction commands

From the archive root, run

```
python -m pip install -r requirements.txt
python code/verify.py -order 120
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact computation uses only the Python standard library. NumPy and SciPy are used for the large- n floating-point envelope tables. The article has no external figure or bibliography-file dependency.

13 A focused further-research program

The following questions extend proved results rather than presupposing unproved summability or novelty claims. Their priority refers to their usefulness for the repository's transseries-inversion program.

13.1 Directional summability of the quadratic model

Determine the growth of $B_1(t)$ on nonpositive rays and in sectors, not only on the positive axis. Is the left-sector solution of Theorem 9.1 the ordinary Borel sum of $\widehat{U}$ in direction π ? A successful proof must establish the required sectorial Borel growth and compare the Laplace sum with the literal fixed point. The coefficient Gevrey bound and Poincaré realization alone do not imply this comparison. Cancellation in $B_1(-t)$ is the central new issue.

Editorial note (ProveIt, 2026-09-29). Later packages in `Analysis/Transseries/docs/series-and-transseries/` answer this question in three senses; none of them has been independently reviewed. (i) *Fine sense, answered positively.* `Negative_Ray_Summation_Exponential_Feedback/` proves (its Theorem `thm:threshold`) that, for these unit amplitudes, $\widehat{U}$ and $\widehat{Q}$ are finely (fixed-width half-strip) Borel–Laplace summable in direction π exactly when $\lambda_j = O((j \log j)^2)$, which includes the quadratic model, with sums that satisfy the literal kernel near the negative axis; its Theorem `thm:borel` continues the inverse Borel transform to $|\zeta| + \Re \zeta < 1/(2A)$, $A = \limsup \lambda_j/(j \log j)^2$. (ii) *Angular sense, answered negatively for $\lambda_j = j^2$.* `Natural_Boundaries_Quadratic_Exponential_Feedback/` proves (its Theorem `thm:main`) that the Borel transform of $\widehat{Q}$ is entire and exponentially bounded on every tube around the negative ray but in no wedge around it, so neither $\widehat{Q}$ nor $\widehat{U}$ is 1-summable in an open arc of directions containing π . (iii) *Orders $k > 1$.* `Sharp_Negative_Direction_Summability_Exponential_Feedback/` (its Theorems `thm:headline` and `thm:ksum`) proves k -summability in direction π , with sum the canonical left-sectorial solution, exactly when $\lambda_j = O((j \log(j+1))^{1+1/k})$; it leaves $k = 1$ open.

13.2 Acceleration and the choice of analytic equation

The law $\log \log B_1(t) \sim 2\sqrt{t}$ suggests that ordinary Laplace summation is not the appropriate positive-direction operation. Develop an accelerated transform adapted to this growth and state explicitly which analytic continuation of the countable kernel it respects. A literal positive-real solution of (1.1) is impossible when λ_j/j is unbounded, so an accelerated value cannot be claimed to solve that divergent literal sum without specifying an additional continuation or regularization rule. Existence, uniqueness, and dependence on that choice are separate research questions.

13.3 Beyond logarithmic coefficient asymptotics

Refine (2.4) to include its sublinear terms and eventually a full multiplicative equivalent. The exact composition entropy, the small nonunit actions, and the shift of the large-action saddle all become relevant below order n . Establish a local limit theorem for K and $J_{\max}$ with a correctly centered fluctuation law. The exponential concentration proved here supplies localization, but not the necessary subleading analysis.

Editorial note (ProveIt, 2026-09-29). Answered for two slope classes by later packages. For $\lambda_j = aj^p$, $p > 1$, after finitely many nonnegative changes, `Finite_Core_Universality_Exponential_Feedback/` proves a multiplicative saddle equivalent with its determinant prefactor (Theorem `thm:saddle` there), a conditional local Gaussian law and an unconditional central limit theorem for the largest action (Theorem `thm:clt`), and an explicit first correction for $p > 3/2$ (Proposition `prop:first-correction`). For $\lambda_j = j^2 + \alpha j + \eta j/\log j + o(j/\log j)$, `Microscopic_Condensation_Exponential_Feedback/` proves a full multiplicative equivalent (Theorem `thm:coefficient`), which for eventually exact j^2 is the $p = 2$, $a = 1$ case of the finite-core first-correction formula, joint total-variation Gaussian–Poisson laws for

the largest action and the number of twos, a two-dimensional local limit, and a central limit theorem for K (Theorem `thm:probability`, Corollaries `cor:local` and `cor:K`). `Poisson_Layers_Finite_Core_Boundary/` adds the first omitted action as an explicit Poisson layer. Slopes $aj^p(\log j)^\beta$ with $\beta \neq 0$ are not treated in these packages.

13.4 The near-linear transition

Analyze slopes such as $j(\log j)^\beta$ with $\beta > 0$ and $jL(j)$ with a slowly varying unbounded L . The present classification already proves divergence together with membership in every positive Gevrey class. What are the sharp coefficient asymptotics and the natural summation scales in this zero-index, nonconvergent regime? The saddle balance changes because the $(p-1)n \log n$ term disappears at $p = 1$.

Editorial note (ProveIt, 2026-09-29). Answered at n -th-root precision by `Near_Linear_Boundary_Exponential_Feedback/` for $\lambda_j = jL(j)$ with L positive, nondecreasing, C^1 and unbounded, of elasticity tending to zero, with $xL(x)$ convex, and with exponentially controlled positive amplitudes: $u_n^{1/n} \sim c_1 e^{b_n}$ where $b_n e^{b_n} = L(n/(1+b_n))$ (its Theorem `thm:main`), with an explicit law for $j(\log j)^\beta$; every factorial-normalized Borel transform is entire but grows too fast for Laplace summation in the positive direction (Theorem `thm:borel`, Corollary `cor:no-laplace`). A multiplicative equivalent for u_n is not claimed there. In the negative direction this class has $A = 0$ in the negative-ray package (fine summability and an entire inverse Borel transform, for unit amplitudes), and the sectorial-summability package gives strong Gevrey- s remainders for every $s > 0$ and k -summability for every $k > 1$.

13.5 Exact type under compositional inversion

For $\widehat{Q} = \widehat{U}^{(-1)}$, determine whether the exact type T_s is preserved and under which hypotheses. We have proved only class membership and equality of the least Gevrey index. A proof of a sharper claim must control cancellations in the reversion formula rather than import the positive-weight argument unchanged.

Editorial note (ProveIt, 2026-09-29). Answered by `Sharp_Weighted_Type_Formal_Reversion/`. Its kernel-independent zero-loss theorem (Theorem `thm:zeroloss` there) shows that tangent-to-identity reversion preserves the weighted type for log-convex weights with $M_n^{1/n} \rightarrow \infty$; combined with (2.2), which it imports from this article and restates but does not re-prove, it gives $T_s(\widehat{Q}) = T_s(\widehat{U})$ for every $s > 0$ (Theorem `thm:inversefeedback`), the inverse Borel radius $1/(4A)$ (Corollary `cor:borelradius`) and a refined inverse type for $\lambda_j \sim aj^p(\log j)^\beta$ (Theorem `thm:refinedfeedback`). Multiplicative inverse asymptotics are proved for $\lambda_j = aj^p$, $p > 2$, in the finite-core package (its Theorem `thm:explicit`); at $p = 2$ that package states a conjecture (`conj:quadratic-inverse`), of which `Quadratic_Exponential_Feedback_After_Reversion/` and `Signed_Quadratic_Feedback_Inversion/` claim independent, not yet reviewed, proofs.

13.6 Amplitudes, general actions, and multiple scales

Replace the unit amplitudes and integer actions by

$$U = \sum_j c_j q^{a_j} e^{\lambda_j U}, \quad c_j \geq 0,$$

first with positive integer a_j and then with well-based real actions. Determine the analogues of the exact type identity in terms of the interaction between c_j , a_j , and λ_j . The proof here uses the availability of arbitrarily many action-one factors in an essential way. For true multilevel transseries, one must also state which filtration is used for formal summability and which parameter controls the analytic normalization. A weight estimate in the wrong filtration is not an analytic certificate.

Editorial note (ProveIt, 2026-09-29). Partly answered for special amplitude classes. The near-linear package allows positive amplitudes with $e^{-D(j-1)} \leq c_j/c_1^j \leq e^{D(j-1)}$. `Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/` treats $U = \beta \sum_j a_j q^j e^{jU}$ with $a_j = a j^{-1-\alpha}$ outside a finite set, $1 < \alpha < 2$, at the critical coupling (a convergent Hahn chart, stable-density sector asymptotics and action budgets); it does not cite this question. The natural-boundaries package relates rational amplitude generating functions with eventually quadratic slopes to angular summability (its Theorem `thm:rational-main`). None of them states an analogue of (2.2) in terms of c_j , a_j and λ_j . The later package `Amplitude_Slope_Compensation_Feedback_Transseries/` (batch 50) does, for integer actions $a_j = j$ and positive exponentially bounded amplitudes: with $d_j = -\log c_j$, the solution converges exactly when $\lambda_j = O(j + d_j)$, is Gevrey- s exactly when $\lambda_j^{1/(s+1)} = O(j \log(j+1) + d_j)$, and, when $d_j/(j \log j) \rightarrow \infty$, has type $T_s(U) = (s+1)^{s+1} \limsup \lambda_j/d_j^{s+1}$ (its Theorems `thm:classification` and `thm:type`; not reviewed). General real actions remain open.

13.7 Signed and vector-valued feedback

For complex amplitudes or slopes, positivity no longer identifies the absolute majorant with the actual coefficient growth. Seek phase or noncancellation conditions under which a sharp lower bound survives. For vector systems, investigate whether a scalar slope budget should be replaced by spectral radii of nonnegative feedback matrices. An adequate theorem should include both a finite-dimensional reduction and a countable-tail hypothesis, rather than silently use a finite-dimensional implicit-function theorem on an infinite system.

13.8 Uniform Gevrey remainder bounds for the sectorial solution

Our left-sector construction has an explicit primitive-action truncation error, while its Poincaré remainder constants are allowed to depend arbitrarily on order. Determine when the actual solution satisfies uniform estimates of the form

$$\left| U(q) - \sum_{n < N} u_n q^n \right| \leq C A^N (N!)^s |q|^N$$

on proper sectors. The sharp coefficient criterion is necessary for such a statement, but its sufficiency for this particular realization requires a new argument. This is the most direct bridge to the Gevrey implicit-function and summability literature.

Editorial note (ProveIt, 2026-09-29). Answered by `Sharp_Negative_Direction_Summability_Exponential_Feedback/` (its Theorem `thm:headline`): for every $s > 0$ the displayed estimate holds on every proper left sector, for the canonical left-sectorial solution that package reconstructs (Proposition `prop:existence`), exactly when $\lambda_j = O((j \log(j+1))^{s+1})$, and likewise for the compositional inverse (Theorem `thm:inverse`). Its constants depend on the sector; no estimate up to the boundary rays is claimed. For $\lambda_j = j^2$ the negative-ray package proves, for $P = e^u Q$, a uniform remainder with an explicit majorant on a closed left half-disc (its Theorem `thm:remainder`).

13.9 Formalization and certified computation

A useful Lean development can be staged without first formalizing all of transseries theory. The finite coefficient formula, coefficientwise monotonicity, the one-large-action lower bound, and the convex composition bound are natural initial targets. A second stage should formalize the logarithmic envelope and its limit estimates. A third stage can address Tonelli, normal convergence, and sectorial contraction.

For computation, replace the illustrative logarithmic tables by directed interval arithmetic, and attach proof objects to verified slope bounds. The positive and left-sector certificates should remain separate in the API. A result should record whether it establishes formal support, coefficient Gevrey growth, an analytic germ, a sectorial solution, a uniform asymptotic expansion, or Borel summability. These are genuinely different outputs.

14 Conclusions

The countable kernel (1.1) is simple enough to admit exact coefficient formulas yet rich enough to distinguish several regularity notions that can otherwise be conflated in transseries inversion. Its formal solution always exists. Linear slope growth is exactly the threshold for ordinary convergence. For every positive Gevrey order, the sharp threshold is instead $(j \log j)^{s+1}$, and the type is given by the explicit constant in (2.2). The threshold survives formal compositional inversion at the level of class membership.

For regularly varying superlinear slopes, the coefficients are generated at leading exponential precision by one large action together with many unit-action factors. This gives both the term of order n in $\log u_n$ and an exponential concentration theorem. In the quadratic case, the Borel transform is entire but too large for positive-direction Laplace summation. Finally, suppression of large slopes in the left half-plane gives a constructive analytic realization for every nonnegative slope sequence, with a uniform action-truncation certificate.

The resulting improvement to an inversion workflow is concrete: formal well-basedness should be accompanied by an explicit slope budget before analytic or Gevrey regularity is asserted. When a positive majorant fails, a direction-sensitive analytic construction may still be available, but its output must not be confused with a convergent Taylor series or a proved Borel sum.

A Dependency map and proof-audit checklist

Result	Main dependencies and boundary
Formal fixed point	Finite coefficient support and q -adic contraction; no analytic assumption.
Positive coefficient formula	Classical formal Lagrange inversion; $k \leq n$ makes specialization finite.
Convergence equivalence	Positivity, Tonelli, and a product-neighborhood analytic argument.
Logarithmic asymptotic	Eventual convexity, one-large-action lower bound, sharp envelope lemma.
Exact Gevrey type	Logarithmic asymptotic for an upper comparison sequence; sparse-subsequence lower bound.
Concentration	Strict concavity of the envelope and a capped convex composition bound.
Entire Borel growth	Positive coefficients and an elementary maximum-term estimate.
Sectorial realization	Suppression for $\Re u < 0$, disk contraction, normal convergence.
Poincaré expansion	Finite-action germs plus the uniform action-tail bound; no uniform Gevrey remainder claim.
Reversion classification	Classical coefficient inversion and factorial bounds; no exact-type transfer claimed.

For independent review, the most important points to check are the localization steps in Lemma 5.1, the retention of the j/n correction in (7.1), and the distinction between primitive-action truncation and Taylor truncation. The constant in (2.2) is sensitive to the second of these points. The positivity assumption is used in the necessity direction of the convergence criterion and in all sharp lower bounds.

B Repository integration note

A natural location for this note is a new research subdirectory adjacent to the consolidated transseries volume, with its own proof-status record. No existing repository source or branch was modified in preparing this archive. The package contains the article source and compiled PDF, a verification script, machine-readable coefficient and envelope tables, and a README explaining reproduction and limitations.

The bibliography distinguishes the repository motivation from external classical tools. The sharp classification theorems are derived within this article; they are not quoted from the repository. The literature check located established infinite-colour inversion and Gevrey implicit function theory, but was not an exhaustive priority search. Independent mathematical review and further bibliographic comparison are required before treating the proposed results as a publication-level novelty claim.

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